Reproducibility Report: Baxter-King, Coppock, Straus and Vavreck (2025)

Endorsements vs. Information: Experimental Evidence of Backlash and Parallel Persuasion during the COVID-19 Public Health Crisis

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Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Baxter-King, Coppock, Straus and Vavreck (2025), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1093/pnasnexus/pgaf185
Replication archive	osf.io/bwv3c
Pre-analysis plans	None; the article states the analyses were not preregistered

The data are not redistributed here. The deposit lives on OSF and that is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` records the OSF file identifier, byte size, MD5 and SHA-256 of each of the 101 deposited files. OSF publishes two hashes of one set of stored bytes, computed by two different algorithms, so both are genuine checks and the script gates on both plus byte size. It also stops if `original/` holds anything the manifest does not list, and stops if a file that is already present no longer matches, which is how a mid-run write into the deposit would be caught rather than silently repaired. Re-hosting a copy would create a second archive that can drift from the first, which is the problem this project exists to document rather than to add to.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it, and `maintained/in_text_claims.R` prints the same numbers again by an independent route. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `baxter-king_etal_2025.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposited archive from OSF, verifies its checksums, fits every model, writes every table and figure into `maintained/output/`, runs the deposited code twice in a throwaway copy, rebuilds the ground truth, prints the in-text claims, and re-verifies the deposit. It takes about ten minutes. Individual scripts can be run on their own, but the three `analysis_*.R` scripts come before the tables and figures that read their model objects.

Required packages: tidyverse, estimatr, patchwork, modelsummary, knitr, kableExtra, scales, openssl, lmtest, here. Paths resolve through `here`, so nothing depends on the working directory. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV, TeX, PNG and RDS output should come back byte-identical. The five PDF figures always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

1 Paper Overview

Citation: Baxter-King, R., Coppock, A., Straus, G. and Vavreck, L. (2025). Endorsements vs. information: experimental evidence of backlash and parallel persuasion during the COVID-19 public health crisis. *PNAS Nexus*, 4(6), pgaf185.

Research question: Which of three kinds of persuasive appeal, endorsements, guidance and mandates, and factual information, move intentions to vaccinate and attitudes to masking, and do their effects differ by party?

Design: Ten survey experiments fielded in the UCLA COVID-19 Health and Politics Project between October 2020 and October 2022, among a total the article reports as 85,191 respondents. Two endorsement experiments randomize the identity of the person endorsing a vaccine; three guidance experiments randomize whether respondents are told the CDC recommends a masking policy or that vaccination is required for an activity; five information experiments randomize factual information about contagiousness and booster effectiveness. All estimates are weighted, use the Lin covariate adjustment, and take HC1 robust standard errors. Effects are reported overall and conditional on seven-point party identification.

Main finding: Endorsements by political figures polarize. Trump's endorsement lowered intentions to vaccinate by 9.2 points on average in October 2020 while raising them among strong Republicans, and by April 2021 endorsements by Biden and Obama lowered intentions among the still unvaccinated in both parties. Guidance and information move Democrats and Republicans in parallel: telling respondents the CDC recommended a masking policy raised support for it by 6 points with no partisan interaction, and information about the bivalent booster raised booster intentions by 8.3 points, again without partisan heterogeneity.

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2 Summary

The ground truth carries 506 rows. Where the deposited code produces a number the article prints, it reproduces it: 445 such comparisons agree at the precision the page prints and 7 do not. The maintained rewrite agrees with the article on 475 values and disagrees on 25, and 6 rows are unverifiable because the article states no number.

2.1 Does the deposited archive run?

The deposit ships 101 files of code, data and metadata and no output directory. Run in order, 61 of its 63 scripts complete and 2 fail. Every write in the deposit goes to a directory the deposit does not ship, so a full run overwrites 0 of the 101 deposited files. Running the archive in place would therefore be safe here, which is not something a reader could know without measuring it.

2.2 Does the maintained rewrite reproduce the paper?

Yes, with two classes of exception. Every coefficient, standard error, sample size and R-squared in Tables 2 and 3 and in appendix Tables S1 to S4 reproduces, as does every estimate and standard error printed on Figures 1, 3 and 5, and every number the article states in prose or in a footnote. Of the 704 numbers printed on the section C appendix figures, 663 reproduce and 41 do not.

The first class is the sample sizes of Table 1, which count respondents enrolled in each experiment while the deposited analysis files hold fewer rows, together with one appendix claim that turns on a coding error in the deposit. The second is new and is a choice this rewrite makes rather than a defect in the article. `lm_lin()` reports the effect at the covariate means, and that quantity exists only while every covariate is interacted with the treatment. In 21 of the 154 conditional cells behind the section C panels, on 11 of the 22 figures, one race category is empty in one of the two arms, which makes the centred race covariate, its treatment interaction, the intercept and the treatment indicator four exactly collinear columns. The fit is the same whichever of the four is dropped and the treatment coefficient is not, so the published number in those cells records which column the decomposition aliased rather than an effect, and no version of any software identifies the quantity there. The rewrite removes the race covariate from those cells and from no others, which restores the estimand where it was missing and leaves the other 133 cells at the covariate list the article used. The `covariates_trimmed` column of `ground_truth/section_c_cells.csv` names the covariate wherever one was removed.

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3 Original Archive Reproducibility

3.1 What runs, and what a stripped copy shows

The deposit is run twice in a throwaway copy, never in place. The first pass runs its scripts in the order `code/main.R` sources them and lets each see what the ones before it wrote. The second runs each script against a copy holding only deposited files, which is the test of whether a script passes only because another script left something behind.

Table 1: Deposited scripts, as shipped and against a copy holding only deposited files

Pass	Errors	Ran clean
as_shipped	2	61
stripped	15	48

13 scripts run in the first pass and fail in the second. All of them read a model object an earlier script wrote, which is the deposit working as intended rather than a defect: `code/main.R` sources the analysis scripts before the tables and figures. It does mean that no table or figure script in this deposit can be checked on its own.

3.2 The two scripts that fail

Table 2: Deposited scripts that fail when the deposit is run in order

Script	Error
code/figures/figure_mini_meta.R	Error in eval(ei, envir) : object 'four_colors' not found
code/figures/appendix_figures_D24_D45.R	Error in 'method(update_ggplot, list(class_any, ggplot2::ggplo

`code/figures/figure_mini_meta.R`, which draws appendix Figure S62, never sources `code/helpers.R` and so cannot find the colour palette it uses. `code/figures/appendix_figures_D24_D45.R`, which draws appendix Figures S24 to S45, composes plots with `+` without loading `patchwork`; under ggplot2 4.x that is an error rather than the silent success it once was. Both are one-line fixes, and both mean that appendix figures the article cites cannot be redrawn from the deposit as it stands.

`code/figures/appendix_figures_E48_E59.R` is a third script worth naming: it exists in the deposit, the deposit's own README describes it, and `code/main.R` never sources it.

3.3 Checksums

All 101 deposited files match the MD5, the SHA-256 and the byte size that OSF publishes for them, and `original/` holds nothing else. The manifest records those published values, so a reader can verify a fresh download against the same numbers; the report does not quote a post-run hash of any file this pipeline writes, because a PDF records the time it was written and its hash changes on every run.

4 Errata

The maintained rewrite corrects one coding error in the deposit. It changes a claim the article makes.

The standard error of a difference in conditional average treatment effects. The deposit's `dic_estimator()` in `code/helpers.R` computes

```
std.error = sqrt(std.error_2 ^2 + std.error_2^2)
```

The second term should be `std.error_1`. As written, the Republican standard error is squared twice and the Democratic one never enters, so every standard error, p-value, confidence interval and equivalence-test p-value in the appendix's equivalence tables is computed from the wrong quantity. This is a typographical error rather than an analytical choice, so the rewrite uses `sqrt(std.error_1^2 + std.error_2^2)`.

The correction moves a published claim, set out as entry 22 of the errata note. The article says that among the four vaccine mandate vignettes "we can affirm equivalence at 10 points in two cases but cannot in two cases." Under the deposit's standard error 2 of the four vignettes affirm equivalence at ten points; under the corrected one, 1 does. The ground truth records this row as a mismatch with `defect_locus = archive`.

The errata note, `errata.qmd`, lists 41 corrections to the article and its appendix, and four items it flags without correcting. Two corrections concern claims the article makes: the equivalence count above, which is entry 22, and the count of cases in which the interaction terms and the differences-in-CATEs disagree in significance, which is four by the printed estimates where the article says one (entry 21). Two more are counts of survey respondents: the abstract, the Significance Statement, and the Introduction give the headline count as 85,191 respondents, where the ten deposited analysis files hold 96,567 distinct respondent identifiers, and the Materials and methods put wave five at 30,000 respondents alongside waves six to eight, where the deposited wave-five files and the appendix's CONSORT diagram hold 14,557. The rest are descriptions, labels, cross-references, wording, spelling, and reference-list details, each quoted from the published page. The note renders to `baxter-king_etal_2025_errata.pdf` at the root of this repository, reading the respondent count and the equivalence count from `maintained/output/` when it is rendered.

Two further defects in the deposit are reported rather than corrected, because correcting them would change the published figures rather than a computation.

The series labels of Figure 2 are swapped. `code/analysis/run_models_w3_endorse_by_arm.R` fits the models named `*_prosocial` on the subset `experiment_arm == "A (Personal)"` and the models named `*_not_prosocial` on `experiment_arm == "B (Social)"`. `code/figures/figure_2.R` then relabels `Prosocial` as `Social` and `Not Prosocial` as

`Personal`. Each framing's estimates are therefore plotted under the other framing's label. The rewrite labels each series by the `experiment_arm` value the model was fitted on.

The rows of the guidance equivalence table are swapped. `code/tables/equiv_test_guidance.R` maps the label `G-1: CDC Mask Guidance 1` to the model object `w7_fit_cdcmask` and `G-3: CDC Mask Guidance 2` to `w6_fit_cdcmask`. G-1 is the June 2021 experiment, whose analysis file is `w6`, and G-3 is the September 2021 experiment, whose file is `w7`. The two rows of the published table carry each other's numbers.

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5 Ground Truth Verification

`ground_truth/build_ground_truth.R` runs as the last step of `run_all.R`. Every published value in it was read from the article or its appendix and is used only as a comparison target; the archive column is read from a fresh run of the deposited code and the rewrite column from `maintained/output/`, so neither can drift from the code that made it. Published values are carried as the digits the page prints, and a value agrees when the pipeline's number, printed to that precision, gives the same digits.

Table 3: Ground truth by location in the article

Table or figure	Rows	Archive agrees	Rewrite agrees	Rewrite differs
Appendix Table S2	80	80	80	0
Appendix Table S3	80	64	80	0
Appendix Table S1	70	70	70	0
Table 3	50	50	50	0
Appendix Table S4	40	32	40	0
Figure 1	32	32	32	0
Figure 3	32	32	32	0
Figure 5	32	32	32	0
Table 2	20	20	20	0
Results, endorsements	14	8	12	1
Table 1	10	0	0	10
Materials and methods	7	2	4	2
Results, information	5	5	5	0
Results, guidance	3	2	2	0
Figure 2	2	0	0	0
Footnote c	2	0	2	0
Abstract and Introduction	1	0	0	1

Appendix Figure S1	1	1	1	0
Appendix Figure S10	1	0	0	1
Appendix Figure S11	1	1	0	1
Appendix Figure S12	1	0	0	1
Appendix Figure S13	1	0	0	1
Appendix Figure S14	1	0	0	1
Appendix Figure S15	1	0	0	1
Appendix Figure S16	1	1	1	0
Appendix Figure S17	1	1	1	0
Appendix Figure S18	1	1	1	0
Appendix Figure S19	1	1	1	0
Appendix Figure S2	1	1	0	1
Appendix Figure S20	1	1	1	0
Appendix Figure S21	1	1	0	1
Appendix Figure S22	1	1	0	1
Appendix Figure S3	1	1	1	0
Appendix Figure S4	1	1	1	0
Appendix Figure S47	1	0	1	0
Appendix Figure S5	1	1	1	0
Appendix Figure S6	1	1	1	0
Appendix Figure S7	1	1	1	0
Appendix Figure S8	1	0	0	1
Appendix Figure S9	1	1	0	1
Figure 4	1	0	0	0
Footnote f	1	0	1	0
Footnote g	1	0	1	0

5.1 Where the rewrite differs from the article

Table 4: Every row where the maintained rewrite disagrees with the article

Claim	Article	Rewrite	Locus
E-1: Vaccine endorsement 1, N	14,946	14906	archive
E-2: Vaccine endorsement 2, N	7,249	7197	archive
G-1: CDC mask guidance 1, N	30,857	30751	archive
G-2: Vaccine mandate vignettes, N	10,298	10275	archive
G-3: CDC mask guidance 2, N	33,088	32943	archive
I-1: Contagiousness conversation, N	8,710	8661	archive
I-2: Delta variant conversation, N	8,710	8692	archive

I-3: Bivalent booster information, N	10,700	10677	archive
I-4: Bivalent booster information (children), N	1,628	1626	archive
I-5: Holiday surge information (children), N	1,715	1711	archive
Survey respondents across the ten experiments	85,191	96567	archive
E-1: Fauci, printed estimates and standard errors reproduced	32	28	rewrite
E-2: Trump, printed estimates and standard errors reproduced	32	30	rewrite
E-2: Trump and Fauci, printed estimates and standard errors reproduced	32	22	rewrite
E-2: Fauci, printed estimates and standard errors reproduced	32	29	rewrite
E-2: Biden and Fauci, printed estimates and standard errors reproduced	32	28	rewrite
E-2: Biden, printed estimates and standard errors reproduced	32	28	rewrite
E-2: Obama, printed estimates and standard errors reproduced	32	28	rewrite
E-2: James, printed estimates and standard errors reproduced	32	30	rewrite
E-2: Ramos, printed estimates and standard errors reproduced	32	30	rewrite
I-4: Bivalent Booster - Child, printed estimates and standard errors reproduced	32	28	rewrite
I-5: Vaccince - Child, printed estimates and standard errors reproduced	32	30	rewrite
E-2 unvaccinated respondents interviewed	7,249	7197	archive
Mandate vignettes affirming equivalence at 10 points	2	1	archive
Experiments where the interaction term and the difference-in-CATEs disagree	1	2	paper_

10 of these are the N column of Table 1, which reports how many respondents were enrolled in each experiment. The deposited analysis file for each experiment holds fewer rows, and the deposit contains no script that assembles Table 1, so nothing in the archive can produce the published figures. One more is the headline count of respondents across the whole project: the union of the respondent identifiers in the ten deposited analysis files is larger than the 85,191 the abstract reports, so the deposit does not contain the file from which that number was counted. Another is the same gap restated in the text for E-2. The remaining 2 are the equivalence claim discussed above and one appendix claim about how often the interaction terms and the differences in conditional average treatment effects agree.

5.2 Coverage of the published floats

The float list is taken from the article and its appendix rather than from what the pipeline happens to produce: 74 floats in all, of which 35 carry at least one ground-truth row.

Table 5: Published floats, by whether the ground truth reaches them

Float group	Floats
Appendix section D, predicted unvaccinated	22
Appendix section D, variable importance	1
Appendix section E, vignette CATEs	14

Meta-analysis of all studies	1
Verified cell by cell	35
Weighted against unweighted	1

Section C of the appendix is the largest block of published numbers in the paper and the easiest to overlook. It draws one two-panel figure per experimental contrast, 22 of them, in the same form as Figures 1, 3 and 5, and each panel prints its estimates and standard errors on the face of the plot. Those 704 numbers are in no table: the appendix regression tables carry one treatment by party interaction per contrast, not eight conditional estimates. `ground_truth/extract_published_appendix_values.R` parses them out of the published PDF by position rather than by reading order, and the parse is checked against Figures 1, 3 and 5, whose values were transcribed independently from rendered pages. 663 of them reproduce, and the 41 that do not are the cells whose published specification is not identified, discussed above.

The 39 uncovered floats are appendix figures, and the largest group is section D, which repeats the section C panels among respondents a machine-learning model predicts will remain unvaccinated. Those panels do print numbers, and nothing here reproduces them: the rewrite does not refit the prediction model that defines the subgroup, and the deposited script that draws them fails. Each uncovered float carries its reason in `ground_truth/float_coverage.csv`, and the build stops if any float has neither a row nor a reason.

5.3 Every number in the article, extracted

`ground_truth/published_claims.csv` is the extraction of the article's prose, its main-text floats and its appendix tables: every numeric token in them, classified by hand. The 704 numbers printed on the faces of the section C appendix figures are extracted separately and in bulk, into `ground_truth/published_appendix_values.csv`, because listing them one by one here would swamp the file without telling a reader anything the per-figure rows above do not.

Table 6: Numeric claims in the article and appendix, by type

Claim type	Claims
pipeline	476
structural	12
descriptive	8
transcribed	6
definitional	3

A `pipeline` or `descriptive` claim must have a row in the ground truth and a block in `maintained/in_text_claims.R`; the build stops if either is missing. `definitional`,

`structural` and `transcribed` claims have no pipeline counterpart by construction: a scale endpoint is checked against the survey instrument, a count of experimental groups against the pipeline object with one row per group, and a number the article takes from another source against that source.

`maintained/in_text_claims.R` prints every one of these numbers beside the article's own sentence, reaching each by its own route from the same committed output the ground truth reads. Two independent derivations that disagree have found something.

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6 Maintained Rewrite

The rewrite replaces 63 deposited scripts with 21 files. The deposit fits each model in its own script and repeats the same forty-line block for every endorser; the rewrite fits all of them by mapping one function over a named list of arms, so a change to the specification happens in one place.

Original	Replacement
`rm(list = ls())`	omitted
`setwd()`	`here::here()` with `.here` at the repository root
`library()` in each script	`source(here::here("maintained", "helpers.R"))`
`coefplot::position_dodgev()`	`position_dodge()` on the discrete y axis
`saveRDS()` repeated once per model	`map()` over the arms, then `walk2()`
Model objects saved with their environment attached	environment dropped, so two runs write identical bytes
`readRDS()` / `saveRDS()`	`read_rds()` / `write_rds()`
`%>%`	`|>`
Tidy summaries saved where the table needs a fitted object	fitted objects saved throughout

Two substitutions are worth naming. `coefplot` has not been updated since 2018 and its `position_dodgev()` is built on ggplot2 3.x internals that emit a warning for every facet under ggplot2 4.x; `position_dodge()` handles a discrete y axis natively and the panels are equivalent. And the deposit saved the mandate vignette models as tidy tables of coefficients, which is why the deposit records no sample size or R-squared for appendix Tables S3 and S4; the rewrite saves the fitted objects, so those cells have a counterpart.

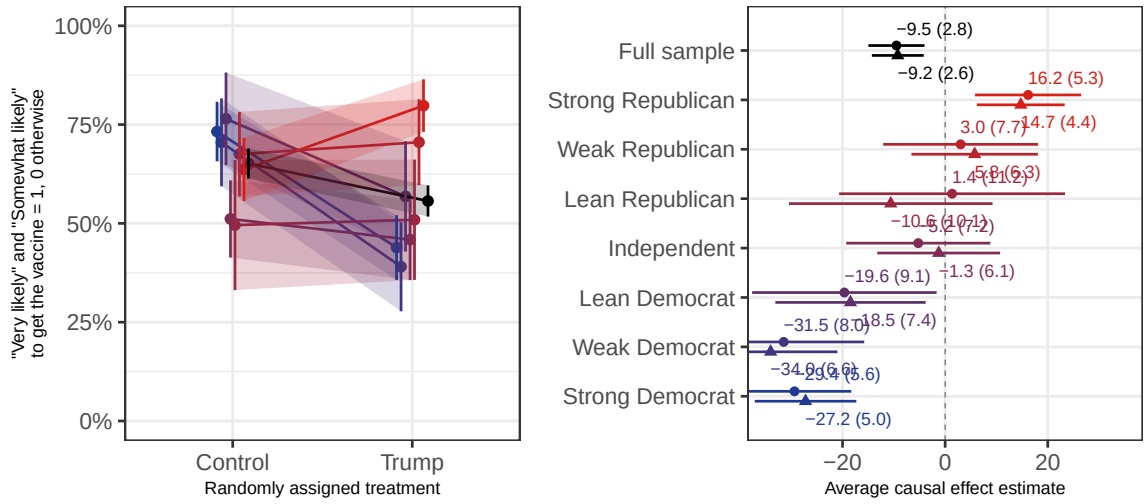
The rewrite also adds four table scripts the deposit lacks or leaves incomplete: appendix Tables S1 to S4, one script each, each writing its cells unrounded to a CSV alongside the formatted LaTeX. Nothing downstream has to read a value back out of a formatted table and round it twice. It adds one more script for the section C appendix figures, which write their estimates to a CSV rather than being redrawn.

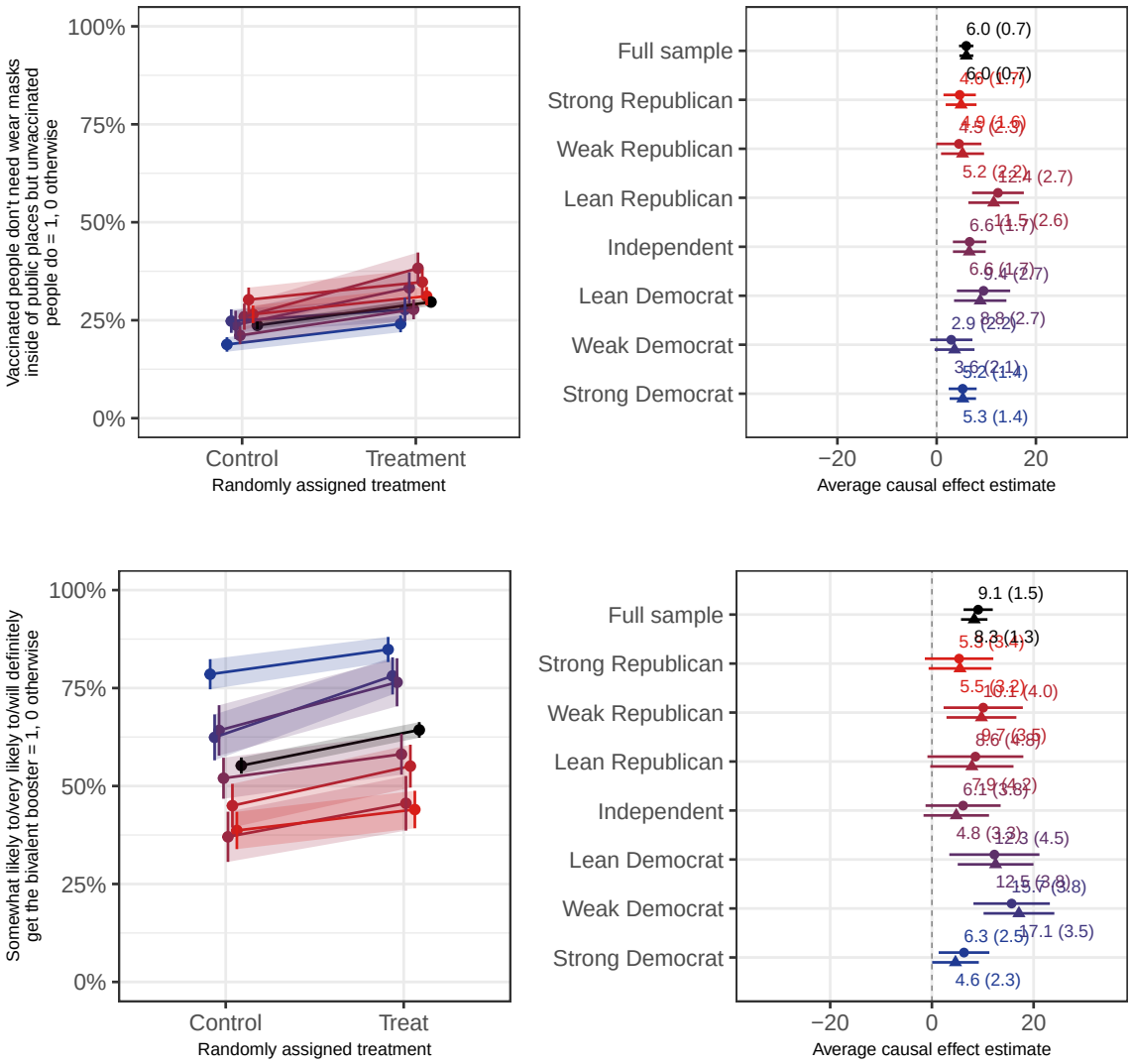
One substitution in the rewrite is worth stating because it is invisible in the deposit's own output. `data_w5_endorse.rds` stores its treatment variable as a factor whose first level is Obama rather than Control, so any contrast taken from that file with the factor as shipped is estimated the other way round: every estimate comes back with the sign reversed while its standard error looks entirely normal. The rewrite sets the control group as the reference before estimating anything, which is a no-op in the other nine analysis files and is what makes the E-2 appendix panels come out with the signs the article prints.

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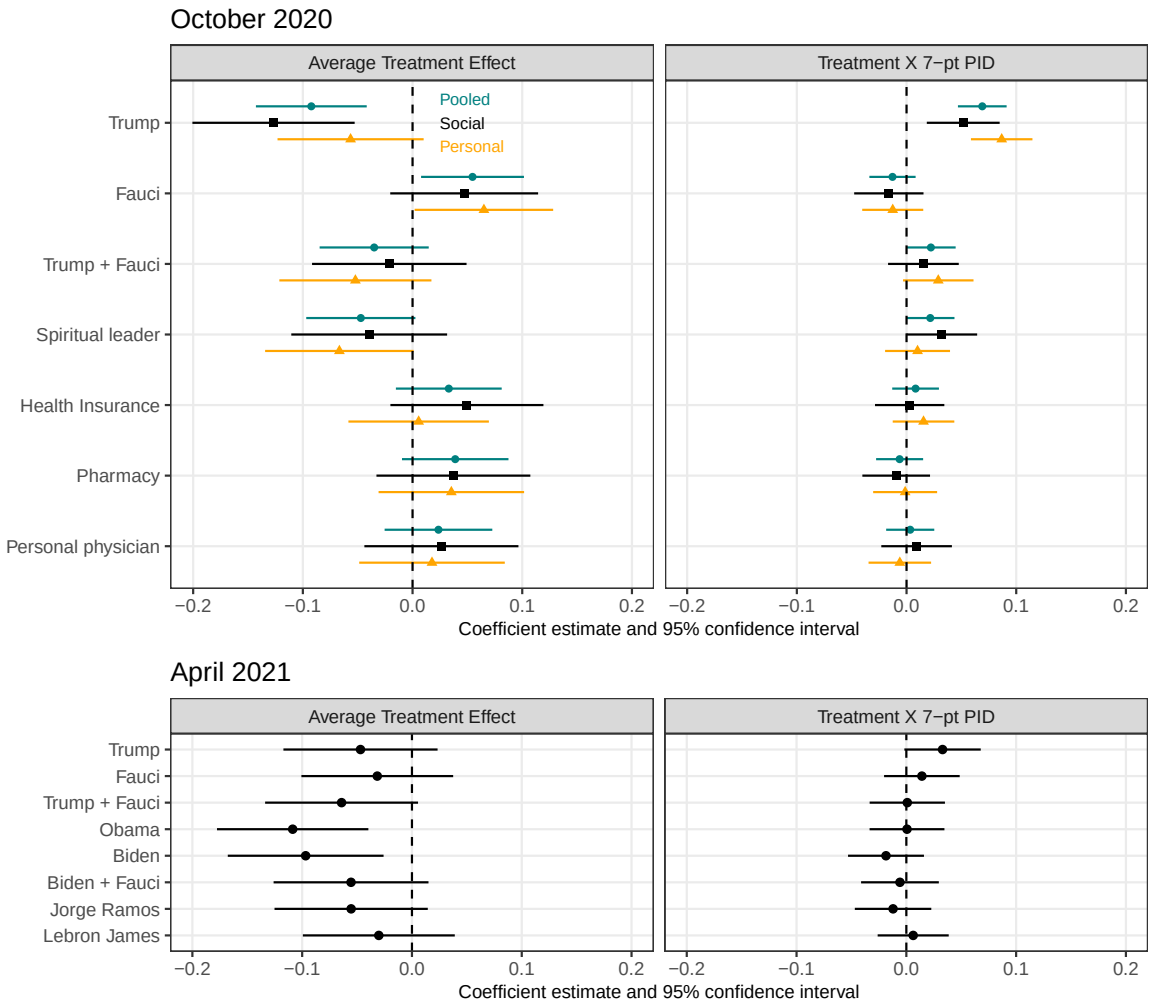
7 Figure Verification

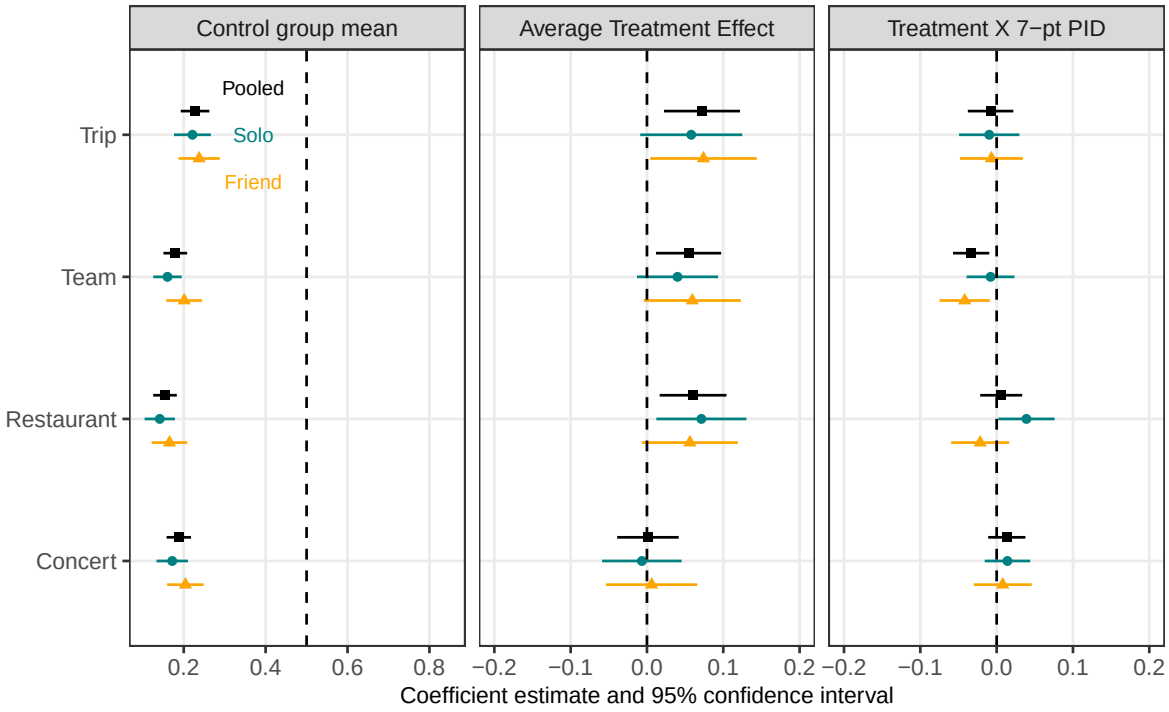
Figures 1, 3 and 5 print every estimate and standard error on the panel, in percentage points. All 96 of those numbers reproduce, as do 663 of the 704 printed on the 22 appendix figures that draw the same panels for the other contrasts.





Figures 2 and 4 print no numbers. Each figure script writes a CSV of every estimate it plots, so the comparison is against the appendix regression tables that print the same coefficients rather than against the panel.


October 2020
Average Treatment Effect
Treatment X 7−pt PID
Pooled
Social
Personal
Trump
Fauci
Trump + Fauci
Spiritual leader
Health Insurance
Pharmacy
Personal physician
−0.2
−0.1
0.0
0.1
0.2
Coefficient estimate and 95% confidence interval
April 2021
Average Treatment Effect
Treatment X 7−pt PID
Trump
Fauci
Trump + Fauci
Obama
Biden
Biden + Fauci
Jorge Ramos
Lebron James
−0.2
−0.1
0.0
0.1
0.2
Coefficient estimate and 95% confidence interval



8 Maintained Rewrite Verification

Nothing in this pipeline draws at random. There is no seed anywhere in the rewrite because there is nothing to seed: every estimate is a weighted least squares fit and every test is analytic. Running `run_all.R` twice from a clean session returns byte-identical CSV, TeX, PNG and RDS output; only the five figure PDFs differ, because a PDF records the time it was written.

The acceptance test for a full run is that `git status --porcelain maintained/output/` shows only those five PDFs.

9 R Environment

Table 7: Package versions this report was built under

Component	Version
R	4.6.1
tidyverse	2.0.0
estimatr	2.0.1.9000
modelsummary	2.6.0
ggplot2	4.0.3
patchwork	1.3.2
kableExtra	1.4.0
here	1.0.2